

A super learner for heterogeneous net treatment effects^a

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Abstract

We introduce a super learner for estimating heterogeneous net treatment effects under unit-varying outcome and cost effects. Our approach is designed for optimal assignment of a binary treatment that induces a cost–benefit trade-off. The net effect and its underlying outcome and cost components are characterized by unknown functional complexity, which our ensemble explores in a data-driven manner.

Directly targeting the net effect performs well when the estimand is simpler than the outcome and cost effects individually. In contrast, separately learning both effects and subsequently aggregating them into the net effect is advantageous when the underlying components share similar functional complexity that translates into a similarly complex target estimand. A hybrid learning strategy succeeds in intermediate settings. Our ensemble nests all approaches and selects the winner by minimizing empirical risk.

In a simulation study, we consider multiple scenarios in which each individual approach dominates the others and show that the ensemble further improves precision across most settings. We additionally present an empirical application using data from a large nonprofit organization, where we analyze the net effect of a fundraising campaign aimed at increasing pledge payments while mitigating donor attrition.

Keywords: Causal inference, cost-benefit analysis, treatment rule, super learner

JEL: C14, C21, D61

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1 Introduction

Combining causal estimands into one target parameter of interest introduces flexibility to translate the target parameter into an estimate. One economically relevant example is the heterogeneous net benefit of a binary treatment, where the treatment variable affects both the outcomes and costs of a unit based on covariates. The relevant net effect directly relies on the underlying outcome and cost effects, which may either be modeled jointly or separately using covariate information. The modeling choice provides a broader scope for misaligned inductive bias — the implicit assumptions an estimator makes about the functional complexity of the target.

More precisely, suppose a decision maker wishes to assign the treatment in a way that maximizes the net benefit across the treated population. Cost efficiency requires the marginal benefit from the treatment, the outcome effect, to at least cover the arising marginal costs, the cost effect. If costs are constant across units, it suffices to estimate CATEs based on the outcome variable and use the constant cost as the cut-off for treatment assignment. However, if costs vary at the individual level, both personalized components must be taken into account, which increases the complexity of the target and, ultimately, the functional complexity one wishes to learn. The key challenge is that the appropriate level of aggregation, whether to target the net effect directly or to estimate its components separately, depends on this unknown structure.

We propose super learning for complex causal estimands as a solution to inductive bias and introduce an ensemble method for estimating the Conditional Net Average Treatment Effect (CNATE). The main contribution of this paper is our proposed method designed to derive unit-level effects in a data-driven manner. We showcase how the underlying

complexity of the response surfaces and the resulting outcome and cost effects shape the complexity of the estimand, which motivates the use of an ensemble. Our approach can further be used for approximating the target population where cost efficiency is satisfied.

We illustrate our method with an empirical example using data from a non-governmental organisation (NGO) that aims to increase pledge donations and targets their donors by fundraising communication. This may have the desired effect of higher payments, but may also serve as a reminder of the active payment the donor plans to cancel and leads to drop out. We show that a targeting mechanism would increase welfare and leverage the targeting operator characteristic to investigate heterogeneity across the population (Yadlowsky, Fleming, Shah, Brunskill, & Wager, 2025).

Our paper relates to three strands of literature. First, it discusses inductive bias in the context of heterogeneous treatment effect estimation. Inductive bias has been discussed by Künzel, Sekhon, Bickel, and Yu (2019), who use different meta-learner algorithms for estimating the CATE. Hahn, Murray, and Carvalho (2020) describe the problem of regularization-induced confounding and bias in CATE estimation, while Caron, Baio, and Manolopoulou (2022) introduce a comprehensive framework comparing different meta-learner strategies. Curth and van der Schaar (2021b) show that standard indirect CATE estimators implicitly bias toward treatment effect heterogeneity, and propose strategies that instead encode an inductive bias favoring shared structure between the potential outcomes. Most recently, Liang, van der Laan, and Alaa (2025) propose a hybrid meta-learner that internally regularizes between direct and indirect learning strategies. Our work extends this discussion by allowing for varying unit-level costs measured in a separate variable.

Second, we use super learning (Dudoit & van der Laan, 2005; Sinisi, Polley, Petersen,

Rhee, & van der Laan, 2007) to combine different learning strategies for CNATE estimation instead of relying on a single procedure. Third, we contribute to the literature on cost-benefit analysis in economics. Zhao and Harinen (2020) model net value optimization using modified meta-learners including the outcome and cost effect. Cost-effectiveness analysis in medicine jointly estimates treatment effects on both costs and health outcomes (Baio, 2012; Bonander & Svensson, 2021; Esser et al., 2025; Gabrio, Baio, & Manca, 2019).

Third, our approach relates to the literature on targeted policy making. Sun, Munro, Kalashnov, Du, and Wager (2024) derive a priority score for targeted assignment using the cost-benefit ratio. In fundraising, recent work derives optimal targeting rules that maximize donations while accounting for campaign costs (Athey, Byambadalai, Cersosimo, Koutout, & Nath, 2024; Breman, 2011; Cagala, Glogowsky, Rincke, & Strittmatter, 2021; Esterzon, Lemmens, & Van den Bergh, 2023), but costs are usually treated as fixed and known. In our empirical application, we analyze how to optimally target donors when both solicitation costs and donation benefits are uncertain and possibly correlated.

This article is organized as follows. Section 2 presents a motivating example that illustrates the role of inductive bias in CNATE estimation. Section 3 introduces the framework and notation. Section 4 formalizes the inductive bias argument. Section 5 introduces the super learner. We analyze finite-sample properties in Section 6 and provide an application in Section 7. Section 8 concludes.

2 Motivating example

Before introducing the formal framework, we illustrate the core challenge of CNATE estimation through a stylized example. Let Y_i be the outcome and C_i the cost of unit i

under a binary treatment $W_i \in \{0, 1\}$. The net effect for unit i is $\psi(X_i) = \tau_Y(X_i) - \tau_C(X_i)$, where τ_Y and τ_C denote the outcome and cost effects. The key question is whether the level of aggregation at which one estimates ψ amplifies or simplifies its functional complexity relative to its components. We consider two extreme cases.

We model the potential outcomes using conditional response surfaces $\mu_{Y(0)}^*(X_i)$ and $\mu_{Y(1)}^*(X_i)$, and the potential costs using $\mu_{C(0)}^*(X_i)$ and $\mu_{C(1)}^*(X_i)$. For simplicity, let the cost in the absence of treatment equal zero ($\mu_{C(0)}^*(X_i) = 0$). Sine functions using a single covariate characterize the remaining response surfaces. The parameter Δ denotes a phase shift, and β_y, β_c the level shift in the outcome and cost effect.

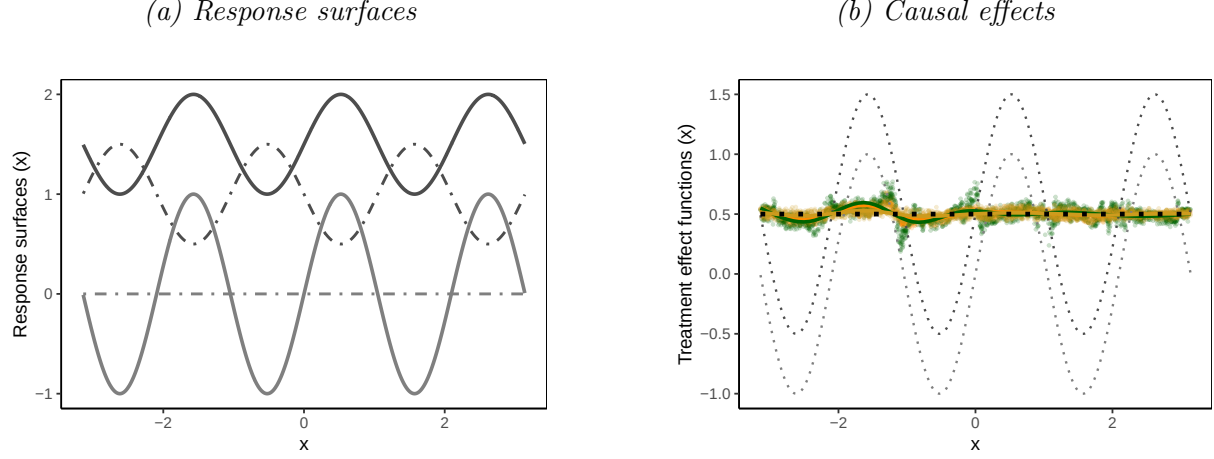
$$\begin{aligned}\mu_{Y(0)}^*(X_i) &= -0.5 \sin(\Delta_{y_0} + \omega_{y_0}x), & \mu_{Y(1)}^*(X_i) &= \beta_y + 0.5 \sin(\Delta_{y_1} + \omega_{y_1}x) \\ \mu_{C(0)}^*(X_i) &= 0, & \mu_{C(1)}^*(X_i) &= \beta_c + \sin(\omega_c x)\end{aligned}\tag{1}$$

Scenario 1: Constant CNATE. Let $\Delta_{y_0} = \Delta_{y_1} = 0$ and $\omega_{y_0} = \omega_{y_1} = \omega_c = 1$. Then $\tau_Y(x)$ and $\tau_C(x)$ are the same function shifted by the constant $\psi(x) = \beta_y - \beta_c$, that precisely equals the net effect.¹ As Figure 1 shows, transforming the outcomes (yellow) or minimizing the loss over separate nuisances jointly (orange) provides a better fit of the true CNATE (black dashed line) compared to separate estimation with subsequent differencing (green).

The reason is that aggregating the response surfaces simplifies the target parameter.

¹An even simpler setting with a constant net effect can be constructed by setting the outcome and cost effects to constants, too.

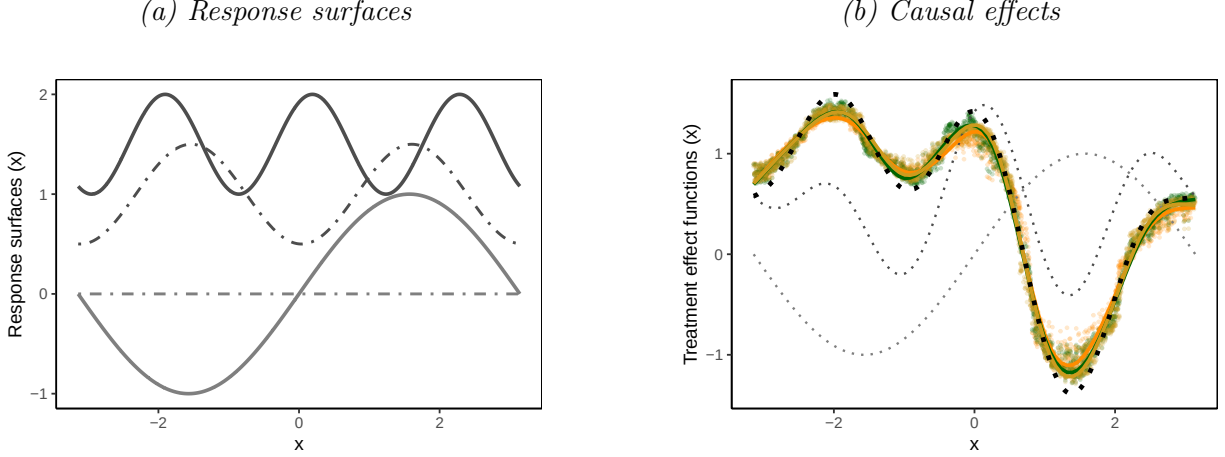
Figure 1: *Motivating example: constant net effect*



Notes: Results for Scenario 1, where the net effect is constant. Sub-figure (a) shows the response surfaces for the potential outcomes, costs and their difference. Sub-figure (b) summarizes the outcome- and cost-effects, the CNATE, and the point predictions under regimes (I)–(III).

Scenario 2: Heterogeneous CNATE. Let $\Delta_{y_0} = 1.5$, $\Delta_{y_1} = 1$, $\Delta_c = 0$, and $\omega_{y_1} = 3$, $\omega_{y_0} = 2$, $\omega_c = 1$. Then $\tau_Y(x)$ and $\tau_C(x)$ become highly complex separately, and even more so when aggregating to the net effect. In this setting, it is advantageous to optimize the fit of all components in isolation. Thus, separately estimating the outcome and cost effects and subsequent differencing yields the best fit for the true CNATE function.

Figure 2: *Motivating example: complex net effect*



Notes: Results for Scenario 2, where the net effect varies across units. Sub-figure (a) shows the response surfaces for the potential outcomes, costs and their difference. Sub-figure (b) summarizes the outcome- and cost-effects, the CNATE, and the point predictions under regimes (I)–(III).

As we do not know the underlying complexity of the target, we propose optimizing over an ensemble that includes all three approaches and a weighted average thereof. This motivates our use of the super learner in Section 5. We formalize the framework in the following section.

3 Framework

Data and setup. We observe n independent and identically distributed tuples $O_i = (X_i, Y_i, C_i, W_i)$, with units $i = 1, \dots, n$ drawn from a distribution P , where $X_i \in \mathcal{X} \subseteq \mathbb{R}^d$ is a covariate vector and $W_i \in \{0, 1\}$ is a binary treatment indicator. Moreover, $Y_i \in \mathbb{R}$ is the outcome of unit i , and $C_i \in \mathbb{R}$ is the cost associated with treatment. Throughout, capital letters denote random variables, lowercase letters denote their realizations. For any function f , we write $f(x)$ when defining the function at a fixed point $x \in \mathcal{X}$, and $f(X_i)$ when evaluating it at the covariates of unit i . We follow the notation in [Nie and Wager](#)

(2020) and use *-superscripts to denote unknown population quantities and hats for their estimates. A function without star or hat denotes a generic candidate being optimized over.

Estimand. Adopting the potential outcomes framework (Neyman, 1923; Rubin, 1974), let $Y_i(w)$ and $C_i(w)$ denote the potential outcomes and potential costs for unit i under treatment $w \in \{0, 1\}$. We define the conditional net average treatment effect (CNATE) as our estimand by

$$\psi^*(x) := \mathbb{E}[\{Y_i(1) - Y_i(0)\} - \{C_i(1) - C_i(0)\} \mid X_i = x], \quad (2)$$

where the expectation is over the common true data-generating distribution P^* of any unit i . Following the notation in M. J. van der Laan and Rose (2011, 2018), we let $\mathcal{M}$ be the statistical model, i.e., a collection of candidate distributions P for O_i . The estimand in (2) is a mapping $\Psi: \mathcal{M} \rightarrow \mathcal{F}$, where $\mathcal{F}$ is a suitable function space on $\mathcal{X}$. For any $P \in \mathcal{M}$, we write $\psi_P(x) = \Psi(P)(x)$ for the CNATE implied by P . Under the true data-generating distribution P^* , this yields $\psi^*(x) \equiv \psi_{P^*}(x) = \Psi(P^*)(x)$.

Identification. Identification of $\psi^*(x)$ from the observed data distribution relies on the following assumptions, formally stated in Assumption 1. First, treatment assignment is unconfounded, meaning that the treatment status is independent of potential outcomes and costs, conditional on observed covariates X_i . Second, the propensity score $e^*(x) = \Pr(W_i = 1 \mid X_i = x)$ is bounded away from zero and one, ensuring that for each value of X_i , comparable units exist in both the treatment and control group. Third, there is no interference between units, so that one individual's treatment assignment does not affect

another individual's outcomes or costs, and the observed outcome and cost for each unit equal the potential outcome and cost corresponding to the treatment actually received.

Assumption 1. *Identifying causal assumptions.*

(a) *Unconfoundedness:* $\{Y_i(0), Y_i(1), C_i(0), C_i(1)\} \perp\!\!\!\perp W_i \mid X_i$.

(b) *Overlap (common support):* There exists $\varepsilon > 0$ such that $\varepsilon < e^*(x) < 1 - \varepsilon$ for all $x \in \mathcal{X}$, where $e^*(x) = \Pr(W_i = 1 \mid X_i = x)$ is the propensity score and $\mathcal{X}$ denotes the support of X_i .

(c) *Stable unit treatment value assumption (SUTVA):* We have consistency of observed and potential outcomes and costs by

$$Y_i = W_i Y_i(1) + (1 - W_i) Y_i(0),$$

$$C_i = W_i C_i(1) + (1 - W_i) C_i(0).$$

Moreover, the potential outcomes $\{Y_i(w), C_i(w)\}_{w \in \{0,1\}}$ depend only on the treatment assignment W_i of unit i itself, and are not affected by the treatment assignments W_j of any other unit $j \neq i$.

Under Assumption 1, the CNATE in (2) is identified from the observed data O_i ; see Appendix A.1. Due to the linearity of the expectation in (2), there exist several estimation strategies that are asymptotically equivalent but differ in finite samples (Liang et al., 2025).

Estimation strategies. We estimate $\psi^*(x)$ by empirical loss minimization of a doubly robust loss function. Throughout, we adopt three variants of the R-loss introduced in Nie

and Wager (2020), summarized in Table 1.² We denote $\nu^*(x) = (\mu_Y^*(x), \mu_C^*(x), e^*(x))$ as a vector of nuisance parameters, where $\mu_Y^*(x) = \mathbb{E}[Y_i \mid X_i = x]$ is the conditional mean outcome and $\mu_C^*(x) = \mathbb{E}[C_i \mid X_i = x]$ is the conditional mean cost.³

The transformed R-learner uses $Z_i = Y_i - C_i$ and $\mu_Z^*(x) = \mathbb{E}[Z_i \mid X_i = x]$ and directly targets $\psi^*(x)$ by minimizing $L_{\text{tf}}(\psi)$. The joint R-learner also targets $\psi^*(x)$ directly but includes nuisances $\mu_Y^*(x)$ and $\mu_C^*(x)$ separately when minimizing $L_{\text{joint}}(\psi)$. The separate R-learner first minimizes two independent loss functions, $L_Y(\tau_Y)$ and $L_C(\tau_C)$, by individually targeting the CATE of outcomes and costs,

$$\tau_Y^*(x) = \mathbb{E}[\{Y_i(1) - Y_i(0)\} \mid X_i = x], \quad (3)$$

$$\tau_C^*(x) = \mathbb{E}[\{C_i(1) - C_i(0)\} \mid X_i = x], \quad (4)$$

before the final estimator takes the difference $\hat{\psi}_{\text{sep}}(x) = \hat{\tau}_Y(x) - \hat{\tau}_C(x)$.⁴ Section 4 contrasts the corresponding estimators based on the different R-learner loss functions in Table 1 in more detail.

4 Inductive bias

The motivating example in Section 2 illustrates that the appropriate estimation strategy depends on the unknown functional complexity of the CNATE and its components. We now formalize this observation. The key question is whether the level of aggregation in Table 1

²Our approach could similarly be implemented using a different doubly robust loss, such as the AIPW-loss. We use the R-loss because it estimates the CNATE in the region with overlap and therefore performs well with observational data, avoiding instability when propensity scores lie close to the boundaries (Foster & Syrgkanis, 2023; Morzywolek, Decruyenaere, & Vansteelandt, 2024).

³For a binary treatment, $e^*(x) = \Pr(W_i = 1 \mid X_i = x) = \mathbb{E}[W_i \mid X_i = x]$.

⁴A similar estimation strategy can be found in Bonander and Svensson (2021).

Table 1: *R-learner loss functions overview.*

<i>Strategy</i>	<i>R-loss</i>
I. Transformed	$L_{\text{tf}}(\psi) = \mathbb{E} \left[\left((Z_i - \mu_Z^*(X_i)) - (W_i - e^*(X_i)) \psi(X_i) \right)^2 \right]$
II. Joint	$L_{\text{joint}}(\psi) = \mathbb{E} \left[\left((Y_i - \mu_Y^*(X_i)) - (C_i - \mu_C^*(X_i)) - (W_i - e^*(X_i)) \psi(X_i) \right)^2 \right]$
III. Separate	$L_Y(\tau_Y) = \mathbb{E} \left[\left((Y_i - \mu_Y^*(X_i)) - (W_i - e^*(X_i)) \tau_Y(X_i) \right)^2 \right]$
	$L_C(\tau_C) = \mathbb{E} \left[\left((C_i - \mu_C^*(X_i)) - (W_i - e^*(X_i)) \tau_C(X_i) \right)^2 \right]$

Notes: Summary of R-learner losses that target the CNATE.

amplifies or cancels the complexity of the estimand in (2). By utilizing the approaches I.–III., we implicitly impose assumptions on how to learn $\psi^*(x)$. This set of assumptions about the underlying complexity can be considered as inductive bias.

Inductive bias in CATE estimation. In CATE estimation based on outcome Y_i only, inductive bias refers to the assumptions an estimator makes about the relationship between conditional response surfaces $\mu_{Y(0)}^*(X_i)$ and $\mu_{Y(1)}^*(X_i)$, whose contrast $\tau_Y^*(x)$ is the estimand but is never observed for any unit (Holland, 1986). Two key axes determine this bias: the degree of structure shared between potential outcomes, ranging from full pooling to full separation (Curth & van der Schaar, 2021b; Hahn et al., 2020; Künzel et al., 2019), and whether one regularizes the potential outcomes indirectly or $\tau_Y^*(x)$ directly. This matters since direct targeting can exploit simpler effect structure (Curth & van der Schaar, 2021a; Kennedy, 2023).

Extension to the CNATE. The discussion on inductive bias for $\tau_Y^*(x)$ directly connects to our analysis of the CNATE and may amplify the issue due to further uncertainty about the cost effect $\tau_C^*(x)$ and the nuisance component $\mu_C^*(x)$. If the assumptions encoded in the

estimator about the complexity of $\psi^*(x)$ and the underlying relationships between $\tau_Y^*(x)$, $\tau_C^*(x)$, $\mu_Y^*(x)$, $\mu_C^*(x)$ deviate from the truth, the estimator may perform poorly in finite samples.

Aggregation level and estimation error. The three strategies in Table 1 span a spectrum of aggregation levels. At one extreme, the transformed R-learner is the most parsimonious approach and collapses the problem immediately by using a single composite outcome Z_i . However, estimating the CNATE in one step forces a single nuisance model to capture the joint structure of outcomes and costs. At the other extreme, the separate R-learner treats outcomes and costs as entirely distinct causal inference problems, estimating each treatment effect with its own tailored nuisance functions before combining them by subtraction. While this offers maximal flexibility, it may accumulate estimation error from two independent learning problems. The joint R-learner occupies a middle ground: it retains separate nuisance models to respect the potentially different signal structures of outcomes and costs while still targeting the net effect directly in a single loss function.

Motivation for the ensemble. In any given application, the functional form of the nuisance functions, the heterogeneity of the causal effects, and the way they combine into the CNATE are all unknown. Each of these unknowns can influence how well an estimator performs in finite samples. Instead of committing to a single estimation strategy upfront, we combine all three learners through a super learner (Dudoit & van der Laan, 2005; Sinisi et al., 2007), which uses cross-validation to determine the optimal weighting of candidates based on their empirical performance. Section 5 embeds the estimation of the CNATE with the R-learners in Table 1 into the super learner framework.

In any given application, the functional form of the nuisance functions, the heterogeneity of the causal effects, and the way they combine into the CNATE are all unknown, and each of these unknowns can materially influence which strategy performs best in finite samples. Section 5 embeds the three R-learners into an ensemble that resolves this uncertainty in a data-driven manner.

5 Super learner

Rather than committing to a single estimation strategy upfront, we nest all three R-learners together with a continuously stacked combination into a super learner library (Dudoit & van der Laan, 2005; Sinisi et al., 2007). Cross-validation then selects the discrete winner across all four candidates based on empirical performance, ensuring that the final estimator inherits the strengths of whichever strategy suits the unknown complexity of the application at hand.

Candidate library. We define a library of $K = 4$ candidate estimators. The first three candidates $\hat{\psi}_{\text{tf}}$, $\hat{\psi}_{\text{joint}}$, and $\hat{\psi}_{\text{sep}}$ correspond to the transformed, joint, and separate R-learners introduced in Section 4. The fourth candidate is a continuous super learner that combines the three base learners through a weighted average,

$$\hat{\psi}_{\text{stack}}(x) = \hat{\alpha}_1 \hat{\psi}_{\text{tf}}(x) + \hat{\alpha}_2 \hat{\psi}_{\text{joint}}(x) + \hat{\alpha}_3 \hat{\psi}_{\text{sep}}(x), \quad (5)$$

where the stacking weights $\hat{\boldsymbol{\alpha}} = (\hat{\alpha}_1, \hat{\alpha}_2, \hat{\alpha}_3)^\top$ are estimated by constrained least squares subject to $\alpha_k \geq 0$ for all k ,

$$\hat{\boldsymbol{\alpha}} = \arg \min_{\boldsymbol{\alpha} \geq 0} \frac{1}{n} \sum_{i=1}^n \left[(\tilde{Z}_i - \tilde{W}_i \boldsymbol{\alpha}^\top \hat{\boldsymbol{\psi}}(X_i))^2 \right], \quad (6)$$

with $\tilde{Z}_i = Z_i - \hat{\mu}_Z(X_i)$, $\tilde{W}_i = W_i - \hat{e}(X_i)$, and $\hat{\boldsymbol{\psi}}(X_i) = (\hat{\psi}_{\text{tf}}(X_i), \hat{\psi}_{\text{joint}}(X_i), \hat{\psi}_{\text{sep}}(X_i))^\top$.

Cross-fitting and empirical risk. To avoid overfitting, all candidate predictions entering both the stacking step and the final model selection are obtained via V -fold cross-fitting. For each fold v , the nuisance functions $\hat{\mu}_Z^{(-v)}$, $\hat{\mu}_Y^{(-v)}$, $\hat{\mu}_C^{(-v)}$, and $\hat{e}^{(-v)}$ are estimated on the complement, and all candidate learners $\hat{\psi}_k^{(-v)}$ are trained on the complement and evaluated on fold v . The stacking weights are likewise estimated fold-by-fold, yielding cross-fitted stack predictions $\hat{\psi}_{\text{stack}}^{(-v)}$. All four candidates are then assessed against empirical risk plugging in the set of nuisance estimates $\hat{\mu}_Y, \hat{\mu}_C, \hat{\mu}_Z, \hat{e}$. The cross-validated empirical risk for candidate k is

$$\hat{R}^{\text{CV}}(\hat{\psi}_k) = \frac{1}{V} \sum_{v=1}^V \frac{1}{n_v} \sum_{i \in \text{Val}(v)} \left[(\tilde{Z}_i^{(-v)} - \tilde{W}_i^{(-v)} \hat{\psi}_k^{(-v)}(X_i))^2 \right], \quad (7)$$

where $\tilde{Z}_i^{(-v)} = Z_i - \hat{\mu}_Z^{(-v)}(X_i)$ and $\tilde{W}_i^{(-v)} = W_i - \hat{e}^{(-v)}(X_i)$.

Winner selection. The discrete super learner selects the candidate with the smallest cross-validated empirical risk,

$$\hat{k}_n = \arg \min_{k \in \{\text{tf}, \text{joint}, \text{sep}, \text{stack}\}} \hat{R}^{\text{CV}}(\hat{\psi}_k), \quad (8)$$

and the final CNATE estimate is $\hat{\psi}^*(x) = \hat{\psi}_{\hat{k}_n}(x)$. The discrete winner is preferred over directly using the continuous stack as the final estimator, since it inherits the theoretical guarantees of the best candidate in the library, whereas the convex combination may not always identify the global minimum of the empirical risk (Dudoit & van der Laan, 2005). The full procedure is summarised in Algorithm 1.

Theoretical guarantees. We derive high-probability regret bounds for the super learner. We assume the following regularity conditions:

Assumption 2 (Regularity conditions).

- (a) **Bounded predictions.** *There exists $M < \infty$ such that $\|\hat{\psi}_k\|_\infty \leq M$ for all $k \in \{1, \dots, K\}$.*
- (b) **Sub-Gaussian noise.** *The residuals $Z_i - \mu_Z^*(X_i)$ and $W_i - e^*(X_i)$ are sub-Gaussian conditional on X_i .*
- (c) **Nuisance convergence rates.** *All nuisance estimators converge at rate $n^{-1/4}$ in $L^2(P)$ on held-out folds: for each fold v ,*

$$\|\hat{\mu}_Z^{(-v)} - \mu_Z^*\|_{L^2} \vee \|\hat{\mu}_Y^{(-v)} - \mu_Y^*\|_{L^2} \vee \|\hat{\mu}_C^{(-v)} - \mu_C^*\|_{L^2} \vee \|\hat{e}^{(-v)} - e^*\|_{L^2} = O_p(n^{-1/4}). \quad (9)$$

Assumption 2(c) is the standard cross-fitting rate condition of Chernozhukov et al. (2018). It is satisfied by a broad class of nonparametric estimators, including random forests and regularised regression, whenever the nuisance functions belong to a sufficiently smooth function class (L. van der Laan, 2026). Assumption 2(a) can be relaxed to a moment condition at the cost of additional notation.

Theorem 1 (Oracle inequality). *Suppose Assumptions 1 and 2 hold. Let $R(\psi) = E[(\tilde{Z}_i - \tilde{W}_i\psi(X_i))^2]$ denote the population R-risk at true nuisances, where $\tilde{Z}_i = Z_i - \mu_Z^*(X_i)$ and $\tilde{W}_i = W_i - e^*(X_i)$. Then the discrete super learner $\hat{k}_n$ defined in (8) satisfies*

$$E[R(\hat{\psi}_{\hat{k}_n})] \leq \min_{k \in \{1, \dots, K\}} E[R(\hat{\psi}_k)] + O(n^{-1/2}). \quad (10)$$

The proof is given in Appendix B.

The $O(n^{-1/2})$ remainder reflects sampling variation across the finite library of $K = 4$ candidates and does not involve nuisance estimation error, which is of strictly smaller order $o(n^{-1/2})$ by the Neyman orthogonality of the R-loss. Because K is fixed, the remainder involves only a $\log K$ factor and requires no complexity regularisation, entropy integral, or localised Rademacher complexity argument that would arise with a growing library (M. J. van der Laan & Dudoit, 2003; van der Vaart & Wellner, 1996).

Algorithm. Algorithm 1 summarizes the estimation procedure.

Algorithm 1 Super learner for the CNATE

Require: Training data $\{(X_i, Y_i, C_i, W_i)\}_{i=1}^n$, number of folds V , candidate library $\{\hat{\psi}_k\}_{k=1}^K$.

- 1: Partition $\{1, \dots, n\}$ into V folds.
 - 2: **for** $v = 1, \dots, V$ **do**
 - 3: Estimate nuisances $\hat{\mu}_Y^{(-v)}, \hat{\mu}_C^{(-v)}, \hat{\mu}_Z^{(-v)}, \hat{e}^{(-v)}$ on fold complement.
 - 4: Fit each base learner $\hat{\psi}_k^{(-v)}$ on fold complement using nuisances from Step 3.
 - 5: Estimate stacking weights $\hat{\alpha}^{(-v)}$ via (6) on fold complement.
 - 6: Form $\hat{\psi}_{\text{stack}}^{(-v)}(X_i) = \hat{\alpha}^{(-v)\top} \hat{\psi}^{(-v)}(X_i)$ for $i \in \text{Val}(v)$.
 - 7: **end for**
 - 8: Compute $\hat{R}^{\text{CV}}(\hat{\psi}_k)$ for all k via (7) using shared nuisances.
 - 9: Select winner $\hat{k}_n$ via (8).
 - 10: **return** $\hat{\psi}^*(x) = \hat{\psi}_{\hat{k}_n}(x)$.
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6 Simulation

We evaluate the finite-sample performance of the three candidate learners and the super learner in a controlled simulation study. The main objective is to show that the super learner reliably identifies the best-performing candidate across settings in which each individual strategy is designed to dominate, and that it does not meaningfully sacrifice precision relative to the oracle-best learner in any scenario.

Implementation. While the super learner framework is agnostic to the choice of base learner, we implement all three strategies using Generalized Random Forests (GRF) (Tibshirani, Athey, & Wager, 2020), which minimize the R-loss by default and are therefore a natural fit for our framework. GRF derives Neyman-orthogonal forest weights by identifying the closest neighbors within the same leaf and combines them in a residual-on-residual regression following Robinson’s partially linear model (Nie & Wager, 2020; Robinson, 1988). The forests are honest in the sense that the subsamples used for growing the trees and for estimating the leaf weights do not overlap, which guards against overfitting. The discrete super learner selects the candidate with the lowest cross-validated empirical risk as defined in (8). We additionally include the continuously stacked ensemble as a candidate in the library, since it can exploit convex combinations of the three strategies that the discrete selector cannot. The stacked weights are estimated by constrained least squares as described in Section 5.

Design. We consider four scenarios that differ in the complexity of the treatment effect functions and the nuisance structure; Table 2 provides the exact specifications. All covariates are drawn independently from $\text{Unif}(0, 1)$, with $p_\tau = 2$ treatment effect covariates, $p_{\text{nuis}} = 2$

nuisance covariates, and $p_{\text{irr}} = 5$ irrelevant noise covariates appended to the design matrix. We set $n \in \{500, 2,000\}$ and draw $R = 100$ independent replications per configuration. Performance is evaluated on a held-out test sample of $n_{\text{test}} = 1,000$ observations drawn from the same distribution. We report the empirical risk and the mean squared error (MSE) of each learner’s predictions against the true $\psi^*(x)$, averaged across replications, as well as the selection frequency of the discrete super learner.

Scenarios. The four scenarios are constructed such that each individual strategy has at least one setting in which it is theoretically favoured, while the super learner is designed to perform well across all of them.

Scenario 1 features an aligned nuisance structure in which $\mu_Z(x) = \mu_Y(x) - \mu_C(x)$ is linear, so estimating the net nuisance directly is easy. The net effect $\psi(x)$ is simpler than either component effect individually, as τ_Y and τ_C share the same nonlinear basis and nearly cancel. This favours Strategy I (Transformed), which collapses the problem into a single composite outcome and directly exploits the simplicity of ψ .

Scenario 2 places the outcome and cost effects on orthogonal covariate subspaces: τ_Y depends only on X_1 and τ_C only on X_2 . Any estimator that pools the two effects — as Strategies I and II do — cannot separately adapt to each subspace. Strategy III (Separate) is the only approach that can exploit this structure by fitting independent forests for each component, making it the natural winner here.

Scenario 3 combines a divergent nuisance structure, where μ_Y and μ_C depend on different nonlinear bases making μ_Z complex, with positively correlated errors ($\rho = 0.8$). The correlated noise reduces the effective variance of the joint residual relative to the sum of two independent residuals, which benefits Strategy II (Joint). By retaining separate

nuisance models for Y and C while still targeting ψ in a single loss, Strategy II avoids the misspecification of Strategy I’s composite nuisance while not accumulating the independent error variance of Strategy III.

Scenario 5 features a homogeneous net effect ($\psi(x) = 0.3$ for all x) despite complex and individually varying τ_Y and τ_C . No single inductive bias is unambiguously favoured: Strategy I benefits from the constant target but struggles with the complex nuisance; Strategy III is penalised by accumulating error from two separate nonlinear problems. This intermediate setting is precisely where the stacked ensemble can add value beyond discrete winner selection.

Results. Table 3 reports the results for $n = 2,000$; the $n = 500$ results confirm the same qualitative patterns at larger variance, consistent with $\sqrt{n}$ -convergence of the nuisance estimators.

In Scenario 1, the Transformed and Joint learners achieve nearly identical empirical risk and MSE, both substantially outperforming Separate (MSE 0.013 and 0.014 versus 0.044). The super learner correctly concentrates selection on Joint and Transformed, picking Joint in 62% of replications. In Scenario 2, Separate wins decisively (MSE 0.032 versus 0.052–0.053 for the remaining learners), and the super learner selects it in 92% of replications, confirming that the empirical risk criterion reliably identifies the orthogonal structure. In Scenario 3, Joint achieves the lowest MSE (0.010) and is selected in 77% of replications at $n = 2,000$, with selection concentrating as the sample size grows and nuisance estimation improves.

Scenario 5 provides the clearest illustration of the ensemble’s added value. The SL stack achieves an MSE of 0.003 at $n = 2,000$, compared to 0.006 for Transformed and 0.014 for Joint, and is selected by the discrete super learner in 62% of replications. This is the only

scenario in which the stacked combination materially outperforms all individual candidates rather than merely recovering the best one, reflecting the intermediate complexity of the target that no single strategy can fully exploit.

Across all scenarios, the SL winner matches or closely approaches the oracle-best individual learner, never being dominated by a substantial margin. The selection frequencies confirm that the empirical risk criterion succeeds in identifying the favoured strategy in each setting, providing empirical support for the theoretical motivation of the ensemble.

Table 2: *Data-generating process specifications*

Sc.	Treatment effects			Nuisance		
	$\tau_Y(x)$	$\tau_C(x)$	$\psi(x)$	$\mu_Y(x)$	$\mu_C(x)$	μ_Z
1	$\phi(x)$	$\phi(x) - \frac{1}{2}(x_1 + x_2) + 0.3$	$\frac{1}{2}(x_1 + x_2) - 0.3$	$\sum_k g_k(x_k) + \ell(x)$	$\sum_k g_k(x_k)$	linear
2	$1.5 + 2 \cdot \mathbf{1}_{x_1 > 0.5}$	$3 \cdot \mathbf{1}_{x_2 > 0.5}$	$\tau_Y - \tau_C$	$\sum_k f_k(x_k)$	$\sum_k h_k(x_k)$	nonlinear
3	$1 + x_1 + x_2$	$1.3 + \frac{1}{2}(x_1 + x_2)$	$\frac{1}{2}(x_1 + x_2) - 0.3$	$\sum_k f_k(x_k)$	$\sum_k h_k(x_k)$	nonlinear
5	$2 \sin 4x_1 + 1.5 \cos 3x_2 + x_1 x_2$	$\tau_Y - 0.3$	0.3	$\sum_k f_k(x_k)$	$\sum_k h_k(x_k)$	nonlinear

Notes: Sc. 1: aligned nuisance, μ_Z linear (favours Strategy I). Sc. 2: orthogonal τ_Y , τ_C on disjoint covariates (favours Strategy III). Sc. 3: divergent nuisance, correlated errors $\rho = 0.8$ (favours Strategy II). Sc. 5: homogeneous ψ , complex components (favours ensemble). ($p_\tau = 2$, $p_{\text{nuis}} = 2$). All covariates $X_k \sim \text{Unif}(0, 1)$. $h_1(x) = 3x^3 - 2x$, $h_2(x) = \cos(2\pi x)$.

Table 3: *Simulation results*

Sim.-par.		Emp. Risk					MSE					Sel. Freq.			
	N	I. tf	II. jt	III. sep	SL stack	SL winner	I. tf	II. jt	III. sep	SL stack	SL winner	I. tf	II. jt	III. sep	SL stack
Sc. 1	500	0.514	0.513	0.523	0.514	0.514	0.029	0.027	0.064	0.030	0.030	22%	62%	2%	14%
	2,000	0.506	0.506	0.513	0.506	0.507	0.013	0.014	0.044	0.012	0.014	37%	62%	0%	1%
Sc. 2	500	1.575	1.578	1.530	1.532	1.531	0.316	0.324	0.126	0.132	0.131	0%	0%	90%	10%
	2,000	1.007	1.007	1.002	1.003	1.002	0.053	0.052	0.032	0.034	0.033	6%	1%	92%	1%
Sc. 3	500	0.981	0.980	0.982	0.983	0.982	0.036	0.033	0.037	0.043	0.037	24%	30%	15%	31%
	2,000	0.482	0.481	0.481	0.481	0.481	0.014	0.010	0.012	0.011	0.011	5%	77%	14%	4%
Sc. 5	500	1.389	1.391	1.411	1.388	1.391	0.021	0.031	0.107	0.017	0.026	27%	31%	2%	40%
	2,000	0.885	0.886	0.894	0.884	0.884	0.006	0.014	0.050	0.003	0.005	29%	9%	0%	62%

Notes: ($p_{\text{nuis}} = 2$, $p_{\text{irr}} = 5$, $R = 100$ replications). Means across replications. Bold marks the lowest value among individual learners per metric and setting. Sel. Freq. is the share of replications in which the super learner selected each individual candidate.

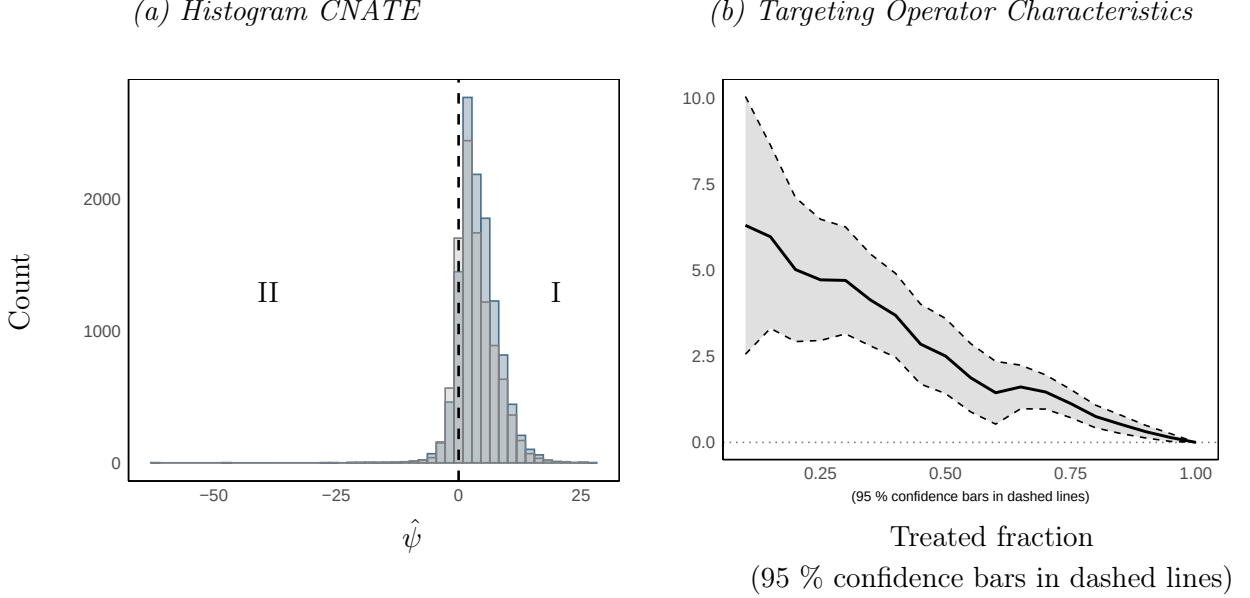
7 Pledge donations

We apply our method to data from a fundraising campaign conducted by a non-profit organization (NGO). The NGO contacts pledge donors by letter, asking them to increase their recurring donations. This intervention could have two opposing effects: first, some donors may respond as intended and increase their payments; second, the mailing may serve as a reminder of their standing order and lead to cancellations. The objective is therefore to maximize donations while minimizing the risk of cancellations through targeted treatment assignment.

Data. The sample consists of $N = 22,038$ pledge donors who contribute on a quarterly or monthly basis. We control for donor characteristics and the past donor journey. The latter captures previous interactions, such as the date and amount of one-time donations or any earlier communication. All donors who receive the mailing have been signed up for at least two years. We use two donor cohorts: the treatment group received the mailing in March 2024, and the control group one month later. The outcome is the increase in pledge donations over the three months following assignment, and the cost variable is the amount of pledge cancellations in the same period. Both are measured in euros.

Estimation. We select the winner among the four candidate learners by minimizing the empirical risk in (8). The empirical risks are 902.07 (I. transformed), 902.32 (II. joint), 902.19 (III. separate), 902.10 (SL stack), selecting the transformed R-learner as the winner. The similarity of the empirical risks across candidates suggests that the net effect is not substantially more complex than its outcome and cost components individually, which favors direct targeting of $\psi^*(x)$.

Figure 3: *CNATE results for fundraising campaign*



Notes: The figure summarizes the results for $\hat{\psi}(x)$. Sub-figure (a) shows the distribution of the CNATE, where W_i corresponds to receiving a letter for increasing the pledge donations, Y_i is the pledge upgrade, and C_i is the pledge cancellations, both measured in euros; sub-figure (b) shows the targeting operator characteristic with 95%-confidence bands based on influence-function standard errors.

We utilize the distribution of estimated net effects to evaluate the impact of the mailing. For inference, we trim observations with estimated propensity scores outside the interval $(0.025, 0.975)$ to ensure stable inverse probability weighting. We then assess overall targeting quality using the rank average treatment effect (Yadlowsky et al., 2025), which summarizes the gain from prioritizing units by their estimated CNATE relative to uniform treatment. The point estimate is 2.833 with a standard error of 0.564, confirming a statistically significant positive effect of the priority score.

Results. The results are graphically summarized in Figure 3. The target group that should receive the mailing is identified by the fraction of donors with a positive CNATE, group I in subfigure 3. This group includes donors for whom the outcome effect outweighs the cost effect. In contrast, the group of donors with a negative CNATE, group II, should

not be targeted. The targeting operator characteristic in subfigure 3 depicts the gain from treating subgroups ranked by their estimated net effect, relative to the average effect from treating the entire sample. The curve is monotonically decreasing and remains significantly above zero throughout, indicating that donors can be meaningfully ranked by their net benefit. For instance, treating only the top 20% of donors by estimated net effect yields a substantially higher average gain than treating the full sample, illustrating the welfare improvements achievable through targeted assignment.

8 Conclusion

We propose a super learner for estimating heterogeneous net treatment effects under unit-varying outcome and cost effects of a binary treatment. The key challenge is that the functional complexity of the net effect and its underlying components is unknown in any given application, and the appropriate level of aggregation, whether to target the net effect directly or to estimate its components separately, depends on this unknown structure. We formalize this challenge as a problem of inductive bias and show that each of the three R-learner strategies we consider can be optimal under a specific data-generating process.

Our simulation study demonstrates this across five scenarios designed so that each individual strategy dominates the others in at least one setting. Directly targeting the net effect via the transformed R-learner performs best when the outcome and cost effects share similar functional form and largely cancel, yielding a simpler net effect. The separate R-learner is advantageous when the outcome and cost effects are driven by orthogonal covariates, so that the net effect is more complex than either component alone. The joint R-learner occupies an intermediate position, retaining separate nuisance models while targeting

the net effect in a single loss. By combining all three strategies through constrained stacking and selecting the discrete winner by minimising cross-validated empirical risk, the ensemble is competitive across all scenarios and strictly dominates any single strategy on average.

In an empirical application using data from a fundraising campaign run by a large non-profit organisation, the transformed R-learner is selected as the winner, with empirical risks across all candidates differing only marginally. This finding suggests that the net effect of the mailing — the difference between pledge upgrades and cancellations — is not substantially more complex than its components individually, so that direct targeting is sufficient. The rank average treatment effect estimate of 2.833 (SE 0.564) confirms that the estimated net effects meaningfully discriminate between donors, and the targeting operator characteristic shows that concentrating the mailing on the highest-ranked donors yields substantially larger gains than uniform targeting.

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A Appendix

A.1 Identification of CNATE

Proof. By linearity of expectation, SUTVA (Assumption 1(c)), and unconfoundedness (Assumption 1(a)):

$$\begin{aligned}
\psi^*(x) &= \mathbb{E}[\{Y_i(1) - Y_i(0)\} - \{C_i(1) - C_i(0)\} \mid X_i = x] \\
&= \mathbb{E}[Y_i(1) \mid X_i = x] - \mathbb{E}[Y_i(0) \mid X_i = x] - \mathbb{E}[C_i(1) \mid X_i = x] + \mathbb{E}[C_i(0) \mid X_i = x] \quad (\text{linearity}) \\
&= \mathbb{E}[Y_i(1) \mid W_i = 1, X_i = x] - \mathbb{E}[Y_i(0) \mid W_i = 0, X_i = x] \\
&\quad - \mathbb{E}[C_i(1) \mid W_i = 1, X_i = x] + \mathbb{E}[C_i(0) \mid W_i = 0, X_i = x] \quad (\text{Assumption 1(a)}) \\
&= \mathbb{E}[Y_i \mid W_i = 1, X_i = x] - \mathbb{E}[Y_i \mid W_i = 0, X_i = x] \\
&\quad - \mathbb{E}[C_i \mid W_i = 1, X_i = x] + \mathbb{E}[C_i \mid W_i = 0, X_i = x], \quad (\text{Assumption 1(c)}) \\
&\tag{A.1}
\end{aligned}$$

where each conditional expectation is well-defined by the overlap assumption (Assumption 1(b)), and the potential outcomes are individual-specific by no interference (Assumption 1(c)). The right-hand side of (A.1) depends only on the distribution of the observed data O_i , so $\psi^*(x)$ is identified. $\square$

A.2 Neyman orthogonality for the score of CNATE

This section shows that Neyman orthogonality holds for our proposed CNATE in (2). Section ?? derives the moment condition based on partially linear regression Robinson (1988). Section A.3 verifies Neyman orthogonality following ?.

Define the partially linear regression models

$$Y = W\tau_{Y,0}(X) + g_{Y,0}(X) + \epsilon_Y, \quad (\text{A.2})$$

$$C = W\tau_{C,0}(X) + g_{C,0}(X) + \epsilon_C, \quad (\text{A.3})$$

and assume $\mathbb{E}[\epsilon_Y | W, X] = \mathbb{E}[\epsilon_C | W, X] = 0$. Under Assumption 1, taking conditional expectations on (W, X) gives

$$\mathbb{E}[Y | W, X] = W\tau_{Y,0}(X) + g_{Y,0}(X), \quad (\text{A.4})$$

$$\mathbb{E}[C | W, X] = W\tau_{C,0}(X) + g_{C,0}(X), \quad (\text{A.5})$$

so that the CNATE in (2) equals

$$\psi_0(X) = \tau_{Y,0}(X) - \tau_{C,0}(X) = \mathbb{E}[Y | W = 1, X] - \mathbb{E}[Y | W = 0, X] - \{\mathbb{E}[C | W = 1, X] - \mathbb{E}[C | W = 0, X]\} \quad (\text{A.6})$$

Let $m_0(X) = \mathbb{E}[Y | X]$, $c_0(X) = \mathbb{E}[C | X]$, and $e_0(X) = \mathbb{E}[W | X]$. Solving for the confounding functions:

$$g_{Y,0}(X) = m_0(X) - e_0(X)\psi_0(X), \quad (\text{A.7})$$

$$g_{C,0}(X) = c_0(X) - e_0(X)\psi_0(X). \quad (\text{A.8})$$

Substituting back yields the residualized equations

$$\underbrace{Y - m_0(X)}_{\tilde{Y}} = \underbrace{(W - e_0(X))}_{\tilde{W}} \tau_{Y,0}(X) + \epsilon_Y, \quad (\text{A.9})$$

$$\underbrace{C - c_0(X)}_{\tilde{C}} = \underbrace{(W - e_0(X))}_{\tilde{W}} \tau_{C,0}(X) + \epsilon_C. \quad (\text{A.10})$$

Subtracting, multiplying by $\tilde{W}$, and taking expectations conditional on X , using $\mathbb{E}[\epsilon_Y \tilde{W} \mid X] = \mathbb{E}[\epsilon_C \tilde{W} \mid X] = 0$, gives

$$\psi_0(X) = \frac{\mathbb{E}[\tilde{Y} \tilde{W} \mid X] - \mathbb{E}[\tilde{C} \tilde{W} \mid X]}{\mathbb{E}[\tilde{W}^2 \mid X]}, \quad (\text{A.11})$$

and the moment condition

$$\mathbb{E}\left[(W - e_0(X))\{(Y - m_0(X)) - (C - c_0(X)) - \psi_0(X)(W - e_0(X))\}\right] = 0. \quad (\text{A.12})$$

A.3 Neyman orthogonality

Let $\nu_0 = (m_0, c_0, e_0)$ denote the true nuisance functions. Let $\mathcal{F}_n$ be the set of all $\nu = (m, c, e)$ consisting of $\mathbb{P}$ -square integrable functions satisfying

$$\|\nu - \nu_0\|_q \leq \text{Const}, \quad (\text{A.13})$$

$$\|\nu - \nu_0\|_2 \leq \delta_n, \quad (\text{A.14})$$

$$\|e(X) - \tfrac{1}{2}\|_\infty \leq \tfrac{1}{2} - \varepsilon, \quad (\text{A.15})$$

$$\|\mu_Y^*(X) - m_0(X)\|_2 \cdot \|e(X) - e_0(X)\|_2 \leq \delta_n n^{-1/2}, \quad (\text{A.16})$$

$$\|c(X) - c_0(X)\|_2 \cdot \|e(X) - e_0(X)\|_2 \leq \delta_n n^{-1/2}. \quad (\text{A.17})$$

We replace (δ_n) by (δ'_n) with $\delta'_n = \text{Const}_\varepsilon \max(\delta_n, n^{-1/2})$.

Assumption 3.1(a). We verify $\mathbb{E}[\psi_{\psi_0, \nu_0}(O)] = 0$. Denote $\mathcal{M}_1 = \mathbb{E}[(W - e_0(X))(Y - m_0(X)) \mid X] - \mathbb{E}[(W - e_0(X))(C - c_0(X))]$ and $\mathcal{M}_2 = \mathbb{E}[\psi_0(X)(W - e_0(X))^2]$. Inserting from (A.9) and using the law of iterated expectations with $\mathbb{E}[\epsilon_Y(W - e_0(X)) \mid X] = \mathbb{E}[\epsilon_C(W - e_0(X)) \mid X] = 0$:

$$\mathcal{M}_1 = \mathbb{E}\left[\underbrace{\tau_{Y,0}(X) - \tau_{C,0}(X)}_{\psi_0(X)}(W - e_0(X))^2\right], \quad (\text{A.18})$$

which cancels $\mathcal{M}_2$, giving $\mathbb{E}[\psi_{\psi_0, \nu_0}(O)] = 0$.

Assumption 3.1(b). The score is linear in $\psi_0(X)$:

$$\psi_{\psi_0, \nu_0}(O) = \underbrace{(W - e_0(X))\{(Y - m_0(X)) - (C - c_0(X))\}}_{\psi_{\nu_0}^A(O)} - \psi_0(X) \underbrace{(W - e_0(X))^2}_{\psi_{\nu_0}^B(O)}. \quad (\text{A.19})$$

Assumptions 3.1(c) and 3.1(d). The map $\nu \rightarrow \mathbb{E}[\psi_{\psi, \nu}(O)]$ is twice continuously Gateaux-differentiable. To verify the orthogonality condition, consider the perturbed nuisances $e_r = e_0 + rh_e$, $m_r = m_0 + rh_Y$, $c_r = c_0 + rh_C$ and differentiate:

$$\left. \frac{d}{dr} \mathbb{E}[\psi_r(O)] \right|_{r=0} = \underbrace{\mathbb{E}[-h_e(X) \cdot \mathbb{E}[-\tilde{Y} + \tilde{C} + \psi_0(X)\tilde{W} \mid X]]}_{\mathcal{T}_1} + \underbrace{\mathbb{E}[\mathbb{E}[W - e_0(X) \mid X] \cdot (-h_Y + h_C + \psi_0 h_e)]}_{\mathcal{T}_2}. \quad (\text{A.20})$$

Both $\mathcal{T}_1 = 0$ and $\mathcal{T}_2 = 0$ follow from $\mathbb{E}[Y - m_0(X) \mid X] = 0$, $\mathbb{E}[C - c_0(X) \mid X] = 0$, and $\mathbb{E}[W - e_0(X) \mid X] = 0$.

Assumption 3.2. The orthogonal moment function is pathwise differentiable and satisfies the Neyman orthogonality condition. Provided nuisance estimators converge at rates faster than $n^{-1/4}$ in mean squared error and under standard regularity conditions (bounded or sub-Gaussian tails, propensity score bounded away from zero and one), the moment function is measurable, square-integrable, and satisfies all smoothness and boundedness requirements of ?.

B Proof of Theorem 1

Notation. Let $O_i = (X_i, Y_i, C_i, W_i)$ and $Z_i = Y_i - C_i$. Write $\eta^* = (\mu_Z^*, e^*)$ for the true nuisances and $\hat{\eta}^{(-v)} = (\hat{\mu}_Z^{(-v)}, \hat{e}^{(-v)})$ for their cross-fitted estimates on fold complement $\mathcal{D}^{(-v)}$.

For any ψ and $\eta = (\mu_Z, e)$, define the R-loss

$$\ell_\eta(O_i, \psi) = \left((Z_i - \mu_Z(X_i)) - (W_i - e(X_i))\psi(X_i) \right)^2.$$

The population risk at true nuisances is $R(\psi) = E[\ell_{\eta^*}(O, \psi)]$, and the cross-validated empirical risk at estimated nuisances is

$$\hat{R}_{CV}(\psi) = \frac{1}{V} \sum_{v=1}^V \frac{1}{n_v} \sum_{i \in \text{Val}(v)} \ell_{\hat{\eta}^{(-v)}}(O_i, \psi).$$

Write $\hat{R}_{CV}(\psi; \eta)$ when the nuisance is held fixed at η .

Basic inequality. Since $\hat{k}_n$ minimises $\hat{R}_{CV}$, for any fixed k^* ,

$$\hat{R}_{CV}(\hat{\psi}_{\hat{k}_n}) \leq \hat{R}_{CV}(\hat{\psi}_{k^*}).$$

Adding and subtracting $R(\hat{\psi}_{\hat{k}_n})$ and $R(\hat{\psi}_{k^*})$ and rearranging gives

$$R(\hat{\psi}_{\hat{k}_n}) - R(\hat{\psi}_{k^*}) \leq \underbrace{[\hat{R}_{CV}(\hat{\psi}_{\hat{k}_n}) - R(\hat{\psi}_{\hat{k}_n})]}_{D_{\hat{k}_n}} - \underbrace{[\hat{R}_{CV}(\hat{\psi}_{k^*}) - R(\hat{\psi}_{k^*})]}_{D_{k^*}}. \quad (\text{B.1})$$

Since $|D_{\hat{k}_n} - D_{k^*}| \leq 2 \max_k |D_k|$, it suffices to show $\max_k |D_k| = O_p(n^{-1/2})$. Decompose each D_k as

$$D_k = \underbrace{[\hat{R}_{CV}(\hat{\psi}_k) - \hat{R}_{CV}(\hat{\psi}_k; \eta^*)]}_{\Delta_k} + \underbrace{[\hat{R}_{CV}(\hat{\psi}_k; \eta^*) - R(\hat{\psi}_k)]}_{\Gamma_k}. \quad (\text{B.2})$$

We bound Δ_k and Γ_k separately.

Nuisance approximation error is second-order. We show $|\Delta_k| = o_p(n^{-1/2})$ for each k . Consider a single fold v and expand $\ell_{\hat{\eta}^{(-v)}}(O_i, \hat{\psi}_k) - \ell_{\eta^*}(O_i, \hat{\psi}_k)$ around η^* . This yields a linear term and a quadratic remainder. Conditional on $\mathcal{D}^{(-v)}$, both $\hat{\eta}^{(-v)}$ and $\hat{\psi}_k^{(-v)}$ are fixed, so the linear term is

$$\frac{1}{n_v} \sum_{i \in \text{Val}(v)} (\tilde{Z}_i^* - \tilde{W}_i^* \hat{\psi}_k^{(-v)}(X_i)) \cdot h(X_i),$$

where h collects the nuisance perturbation $(\hat{\mu}_Z^{(-v)} - \mu_Z^*, \hat{e}^{(-v)} - e^*)$. Taking expectations over $\text{Val}(v)$ conditional on $\mathcal{D}^{(-v)}$, this term vanishes by Neyman orthogonality of the R-loss (Appendix A.2):

$$E[\tilde{Z}_i^* \cdot h(X_i) \mid X_i] = 0, \quad E[\tilde{W}_i^* \cdot h(X_i) \mid X_i] = 0,$$

which follow from $E[Z_i - \mu_Z^*(X_i) \mid X_i] = 0$ and $E[W_i - e^*(X_i) \mid X_i] = 0$. This is the first-order insensitivity condition of Chernozhukov et al. (2018).⁵

The remaining quadratic remainder satisfies

$$|\Delta_k| \lesssim \|\hat{\mu}_Z^{(-v)} - \mu_Z^*\|_{L^2}^2 + \|\hat{e}^{(-v)} - e^*\|_{L^2}^2 + \|\hat{\mu}_Z^{(-v)} - \mu_Z^*\|_{L^2} \|\hat{e}^{(-v)} - e^*\|_{L^2}, \quad (\text{B.3})$$

where the cross-product term arises from the product structure of the R-loss. Under Assumption 2(c) each term is $O_p(n^{-1/2})$, so $|\Delta_k| = o_p(n^{-1/2})$ for all k .

The same argument applies to Strategies II and III, which use $(\hat{\mu}_Y^{(-v)} - \hat{\mu}_C^{(-v)}, \hat{e}^{(-v)})$ as nuisances: the conditions $E[Y_i - \mu_Y^*(X_i) \mid X_i] = 0$ and $E[C_i - \mu_C^*(X_i) \mid X_i] = 0$ kill the linear term, and the quadratic remainder is $O_p(n^{-1/2})$ by the same rate conditions. Since $\mu_Z^* = \mu_Y^* - \mu_C^*$ at the truth, both nuisance configurations target the same population object.

Statistical fluctuation is $O_p(n^{-1/2})$. It remains to bound Γ_k , the centred deviation of the empirical risk from its expectation at true nuisances.

Within-fold concentration. For each fold v , conditional on $\mathcal{D}^{(-v)}$, the terms $\ell_{\eta^*}(O_i, \hat{\psi}_k^{(-v)})$ for $i \in \text{Val}(v)$ are i.i.d. and sub-exponential under Assumptions 2(a)–(b). A Bernstein inequality (L. van der Laan, 2026, Lemma 2) gives, for any $t > 0$,

$$P\left(|\Gamma_k^{(v)}| > t\right) \leq 2 \exp\left(-\frac{n_v t^2}{2(\sigma_k^2 + Mt/3)}\right),$$

where $\sigma_k^2 = \text{Var}(\ell_{\eta^*}(O, \hat{\psi}_k))$. Averaging over folds by the independence of validation sets

⁵Neyman orthogonality of the R-loss is equivalent to pathwise differentiability of the CNATE functional under a local product structure condition in the sense of Chen, Kennedy, and Balakrishnan (2026). The product structure is satisfied here because nuisance errors enter multiplicatively through $(W_i - e(X_i))\psi(X_i)$; see L. van der Laan (2026), Condition B3.

conditional on training complements yields

$$P(|\Gamma_k| > t) \leq 2 \exp\left(-\frac{n t^2}{C_k}\right) \quad (\text{B.4})$$

for a constant $C_k > 0$.

Union bound over K candidates. Since $K = 4$ is fixed, a union bound over candidates gives

$$P\left(\max_{k=1,\dots,K} |\Gamma_k| > t\right) \leq 2K \exp\left(-\frac{n t^2}{C}\right), \quad C = \max_k C_k.$$

Setting $t_n = \sqrt{C \log(2K)/n}$ yields $\max_k |\Gamma_k| = O_p(\sqrt{\log K/n}) = O_p(n^{-1/2})$, since K is fixed.

Assembling the bound. Setting $k^* = \arg \min_k R(\hat{\psi}_k)$ and combining (B.1)–(B.4):

$$R(\hat{\psi}_{\hat{k}_n}) - \min_k R(\hat{\psi}_k) \leq 2 \max_k |\Delta_k| + 2 \max_k |\Gamma_k| = o_p(n^{-1/2}) + O_p(n^{-1/2}) = O_p(n^{-1/2}).$$

Taking expectations under Assumptions 2(a)–(b) (which ensure the O_p term has a finite second moment) converts this to the $O(n^{-1/2})$ bound in (10). $\square$

Extension: the stacked candidate. The proof above covers the three base learners $\hat{\psi}_{tf}$, $\hat{\psi}_{jt}$, and $\hat{\psi}_{sep}$ directly, since each is a fixed function of $\mathcal{D}^{(-v)}$ for a given fold v . We now show that the bound extends to the stacked candidate $\hat{\psi}_{stack}$.

Recall from (5) that for fold v ,

$$\hat{\psi}_{stack}^{(-v)}(X_i) = \hat{\alpha}^{(-v)\top} \hat{\psi}^{(-v)}(X_i), \quad i \in \text{Val}(v),$$

where $\hat{\alpha}^{(-v)}$ solves the constrained least squares problem (6) on $\mathcal{D}^{(-v)}$ and $\hat{\psi}^{(-v)} = (\hat{\psi}_{tf}^{(-v)}, \hat{\psi}_{jt}^{(-v)}, \hat{\psi}_{sep}^{(-v)})^\top$. Since $\hat{\alpha}^{(-v)}$ is a deterministic function of $\mathcal{D}^{(-v)}$, it is fixed conditional on the training complement, and the same conditional independence argument as in Steps 2 and 3 applies to $\hat{\psi}_{stack}$.

The only additional term relative to the base learners is the error induced by estimating $\hat{\alpha}^{(-v)}$ rather than using population-optimal weights α^* . Write

$$\hat{\psi}_{stack}^{(-v)} = \alpha^{*\top} \hat{\psi}^{(-v)} + (\hat{\alpha}^{(-v)} - \alpha^*)^\top \hat{\psi}^{(-v)}.$$

The first term is a fixed linear combination of base learner predictions and is covered by the base learner analysis. For the second term, the induced excess risk satisfies

$$|R(\hat{\psi}_{stack}^{(-v)}) - R(\alpha^{*\top} \hat{\psi}^{(-v)})| \lesssim \|\hat{\alpha}^{(-v)} - \alpha^*\|_2 \cdot E \left[\tilde{W}_i^2 \|\hat{\psi}^{(-v)}(X_i)\|_2^2 \right]^{1/2}. \quad (\text{B.5})$$

Under the boundedness condition in Assumption 2(a), the second factor is $O(1)$. For the first factor, the constrained least squares objective in (6) is strongly convex on the simplex under Assumption 2(b), so standard M -estimation theory gives

$$\|\hat{\alpha}^{(-v)} - \alpha^*\|_2 = O_p(n^{-1/2}).$$

Substituting into (B.5) shows that the additional term is $O_p(n^{-1/2})$, matching the order of Γ_k for the base learners. Hence (10) holds for the full library $k \in \{tf, jt, sep, stack\}$.

C Inference

We quantify uncertainty via the bootstrap of little bags. Under Assumptions 1–6 and Specification 1 in [Tibshirani et al. \(2020\)](#), the GRF is asymptotically Gaussian, allowing inference on $\psi(x)$. The variance is

$$\sigma_n^2(x) := \xi^\top \hat{V}(x)^{-1} H_n(x; \psi(x), \nu^*(x)) (V(x)^{-1})^\top \xi, \quad (\text{C.1})$$

where $\xi^\top = (1 \ 0)$.

The curvature parameter, following Assumption 2 of [Tibshirani et al. \(2020\)](#), is

$$V(x) = \mathbb{E}[(W_i - e^*(x))^2 \mid X_i = x]. \quad (\text{C.2})$$

The inner variance term captures variation from estimating forest weights:

$$H_n(x; \theta, \nu) = \text{Var} \left[\sum_{i=1}^n \alpha_i(x) \psi_{\theta, \nu}(O_i) \right]. \quad (\text{C.3})$$

We approximate (C.3) by the bootstrap of little bags, decomposing into between-group and within-group variation. For large B , the estimator $\hat{H}_n^{BLB}$ converges to the half-sampling estimator of [Efron \(1982\)](#) (see Theorem 6 in [Tibshirani et al. 2020](#)):

$$\hat{H}_n^{HS}(x) = \mathbb{E}_{ss} \left[\left(\frac{1}{\ell} \sum_{b=1}^{\ell} \Psi_b - \Psi \right)^2 \right] - \frac{1}{\ell - 1} \mathbb{E}_{ss} \left[\frac{1}{\ell} \sum_{b=1}^{\ell} \left(\Psi_b - \frac{1}{\ell} \sum_{b=1}^{\ell} \Psi_b \right)^2 \right]. \quad (\text{C.4})$$

In practice this is estimated by

$$\hat{H}_n^{BLB}(x) = \hat{H}_{n,\mathcal{A}}(x) - \frac{1}{\ell - 1} \hat{H}_{n,\mathcal{W}}(x), \quad (\text{C.5})$$

where the across-group variation is

$$\hat{H}_{n,\mathcal{A}}(x) = \frac{1}{G} \sum_{g=1}^G (\bar{\theta}_g(x) - \bar{\theta}(x))^2, \quad (\text{C.6})$$

with $\bar{\theta}_g(x) = \frac{1}{\ell} \sum_{b \in g} \hat{\theta}^b(x)$ and $\bar{\theta}(x) = \frac{1}{B} \sum_{b=1}^B \hat{\theta}^b(x)$, and the within-group variation is

$$\hat{H}_{n,\mathcal{W}}(x) = \frac{G}{\ell - 1} \frac{1}{B} \sum_{g=1}^G \sum_{b \in g} \left(\hat{\theta}^b(x) - \bar{\theta}_g(x) \right)^2. \quad (\text{C.7})$$

D Score functions for local effect estimators

Let $\mathcal{I}_P = \{i : X_i \in P\}$ denote the index set for parent node P , with $n_P = |\mathcal{I}_P|$. Define centered variables $\tilde{Y}_i = Y_i - \bar{Y}_P$ and $\tilde{W}_i = W_i - \bar{W}_P$. The local treatment effect estimator β_P solves

$$\beta_P = \arg \min_b \sum_{i \in \mathcal{I}_P} (\tilde{Y}_i - b \tilde{W}_i)^2, \quad (\text{D.1})$$

with first-order condition $\sum_{i \in \mathcal{I}_P} \tilde{W}_i (\tilde{Y}_i - \beta_P \tilde{W}_i) = 0$. The score function is the summand in this condition:

$$\mathcal{S}_i(\beta_P) = ((Y_i - \bar{Y}_P) - \beta_P (W_i - \bar{W}_P))(W_i - \bar{W}_P). \quad (\text{D.2})$$

By identical arguments for the cost equation:

$$\mathcal{S}_i(\delta_P) = ((C_i - \bar{C}_P) - \delta_P(W_i - \bar{W}_P))(W_i - \bar{W}_P). \quad (\text{D.3})$$

E Pseudo-outcomes

Following the gradient tree algorithm in Section 2.3 of ?, for each child node CN and parent node P define

$$\tilde{\theta}_{CN} = \hat{\theta}_P - \frac{1}{|\{i : X_i \in CN\}|} \sum_{i: X_i \in CN} \xi^\top A_P^{-1} \psi_{\hat{\theta}_P, \hat{\nu}_P}(O_i), \quad (\text{E.1})$$

where $\hat{\theta}_P$ and $\hat{\nu}_P$ are obtained by solving (A.12) in the parent node, ξ selects the θ -coordinate from (θ, ν) , and A_P is a consistent estimate of the gradient:

$$A_P = \frac{1}{|\{i : X_i \in P\}|} \sum_{i: X_i \in P} \nabla \psi_{\hat{\theta}_P, \hat{\nu}_P}(O_i). \quad (\text{E.2})$$

The pseudo-outcomes used for tree splitting are then

$$\rho_i = -\xi^\top A_P^{-1} \psi_{\hat{\theta}_P, \hat{\nu}_P}(O_i) \in \mathbb{R}. \quad (\text{E.3})$$